Pfsense Firewall configuration

Name: Muhammad Yaseen

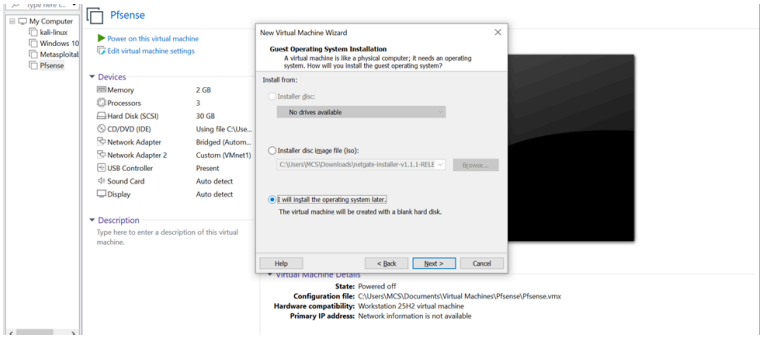
CNIC: 1610224137997

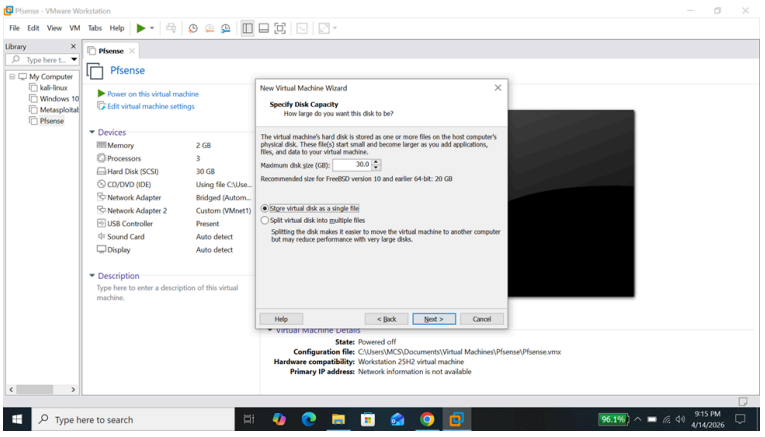
Assignment: Pfsense Firewall Installation & Configuration

▼ VMware & Network Adapter Setting

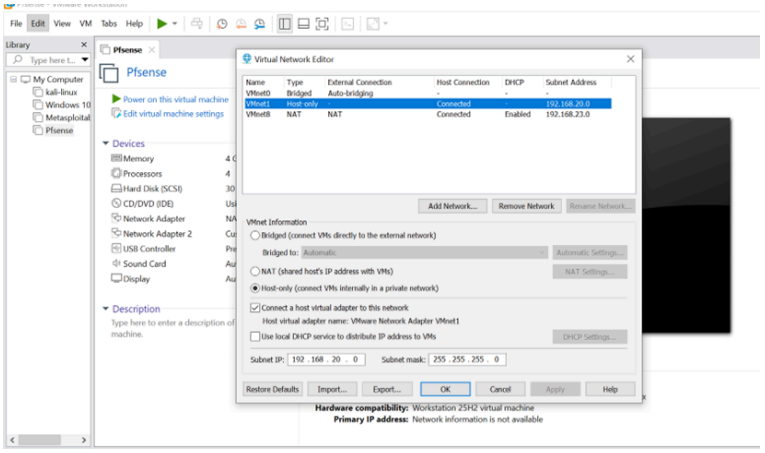
- Create New Virtual Machine
- Your Operating System for this machine should be
  - Other > BSD version 10 64 bit.
- Choose **Typical** (recommended).
- Choose your ISO image.
- Next ...





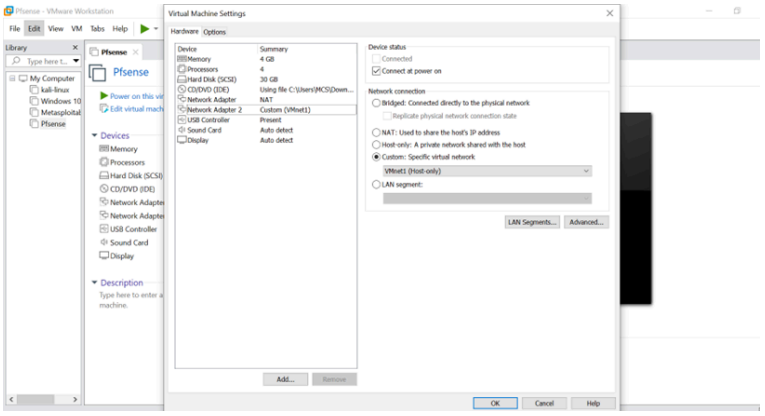


- **Virtual Network Adapter Setting**
- VMware > Edit > Virtual Network Editor > change setting (editable mode).
- Add Network > select a network to add > (choose any b/w VMnet 2-19) > Ok
- Disable DHCP.



- Goto Virtual Machine setting (Pfsense)
- Add > Network Adapter > Finish.

- Set Network Adapter 1 → NAT / Bridged.
- Set Network Adapter 2 → Custom (in my case VMnet1 (Host Only) You can choose your configured.)



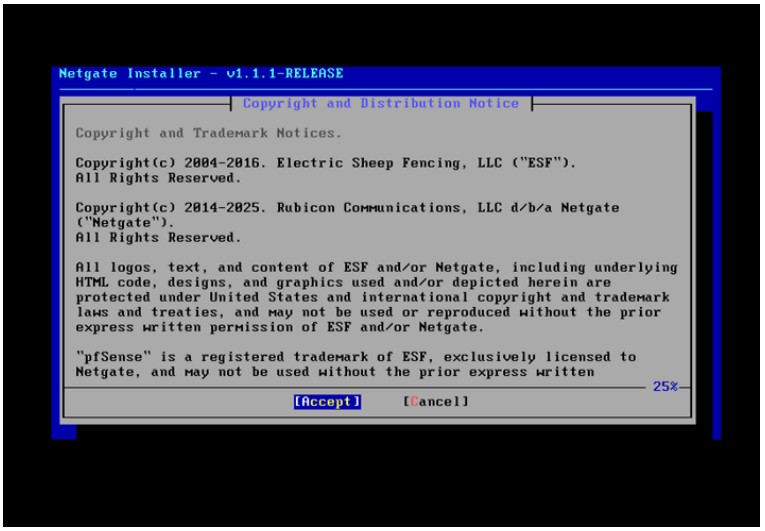
- Click Ok >

▼ Installation

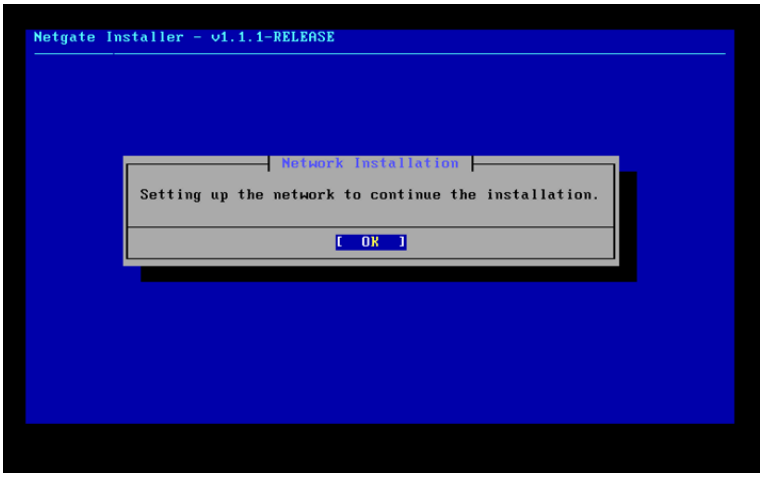
- Start the machine.



- Window will appear
- Copyright and Distribution Notice > Accept

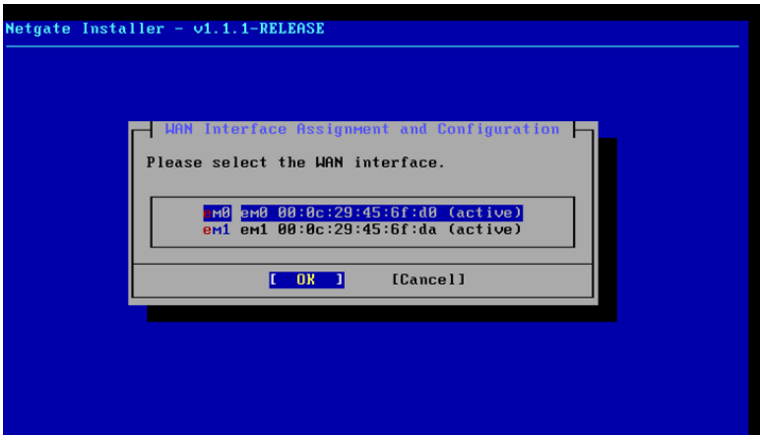


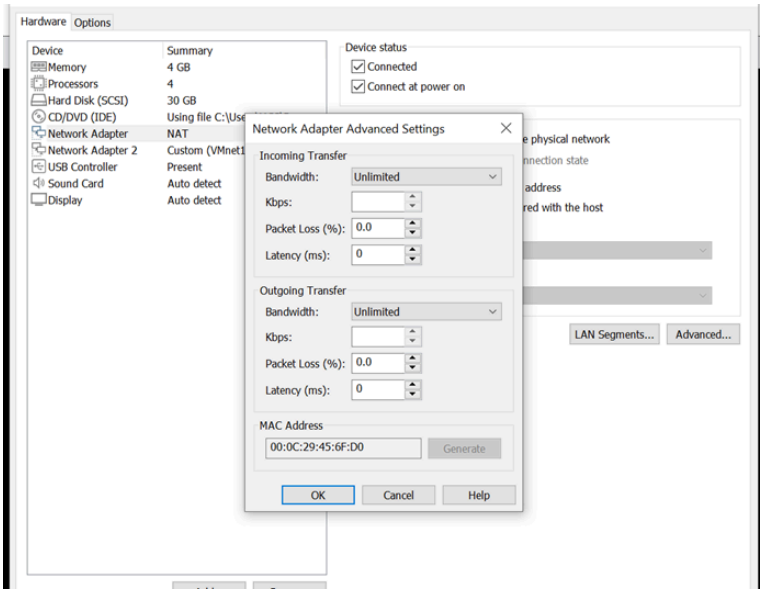
- Network Installation > Ok



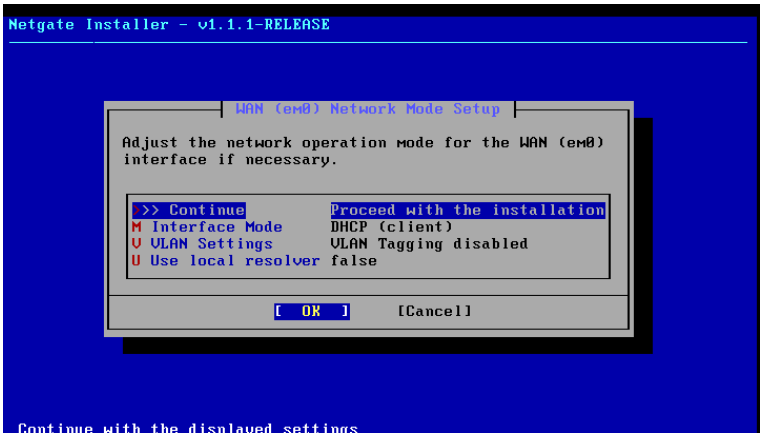
- Wan Interface Assignment and configuration
- For WAN Interface > em0

  Verify the Mac address of interface.

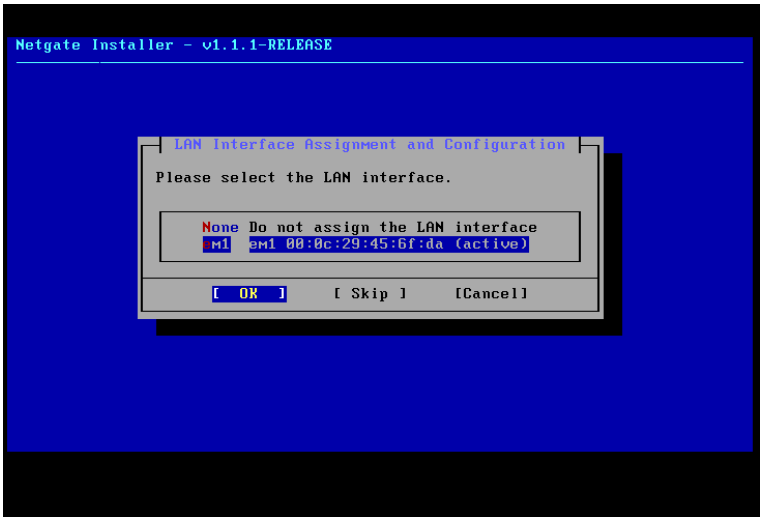


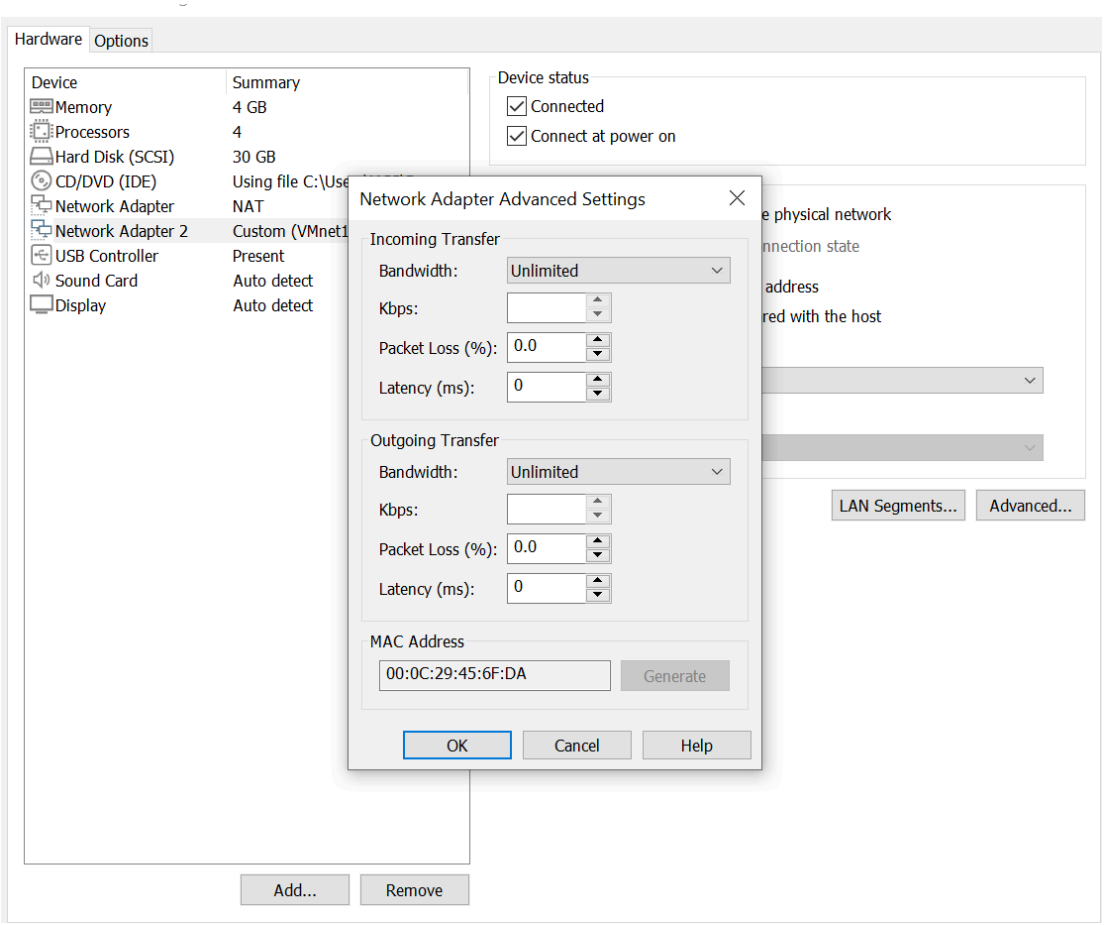


- WAN (em0) Network Mode setup
- Ok >> Continue → Proceed with the installation

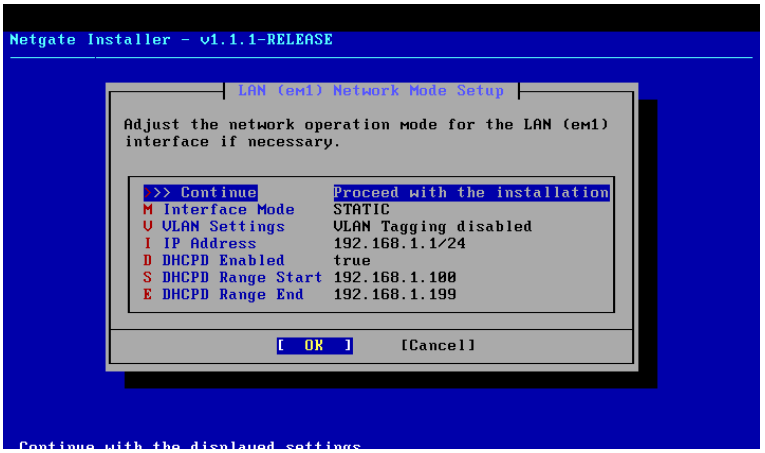


- LAN Interface assignment
- em1 → verify Mac address from

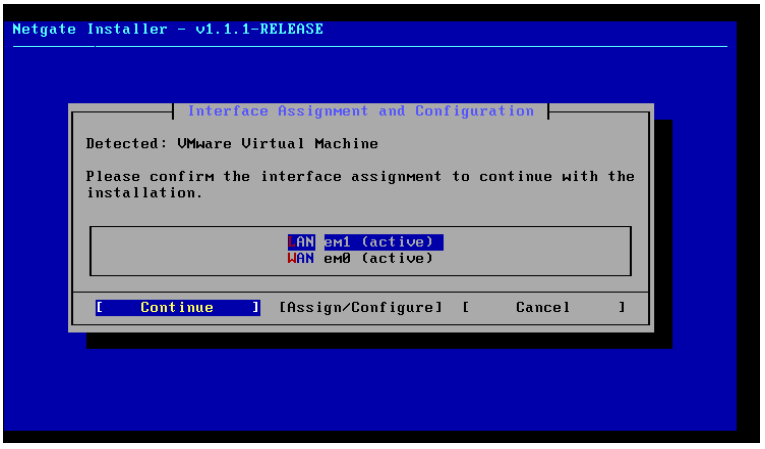




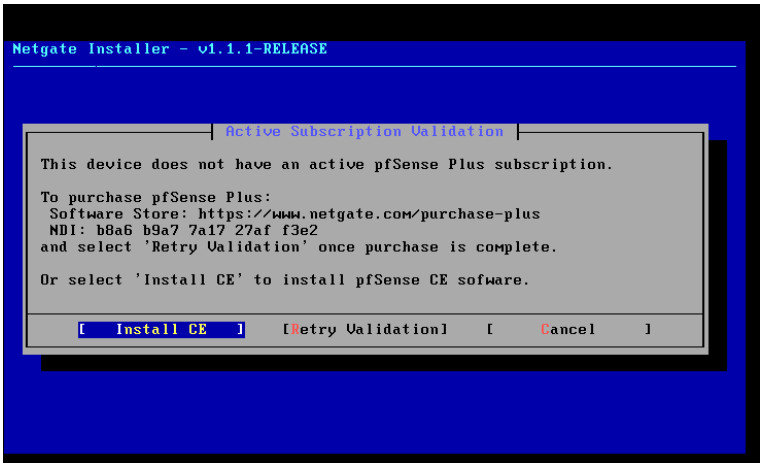
- LAN (em1) Network Mode Setup >> Continue >>OK



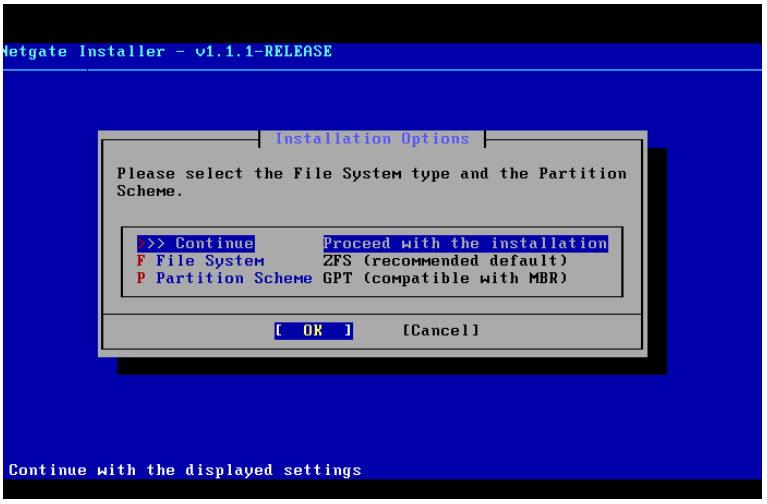
- Interface Assignment & Configuration
- Both are Active. means we are good.
- >> Continue



- Active Subscription Validation >> Install CE



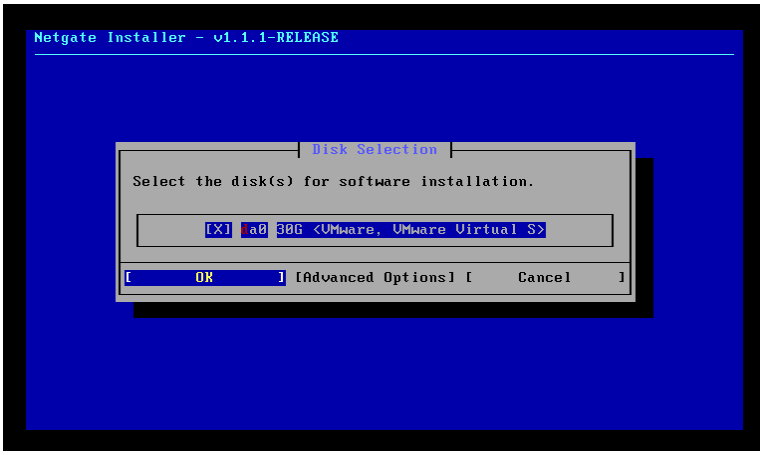
- Installation Options >> Continue (Default is ZFS) Select ZFS if your is not that.



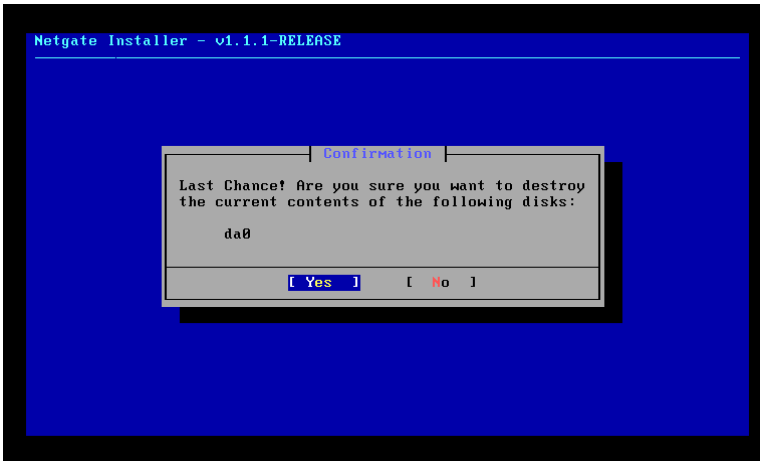
- ZFS virtual Device Type Configuration
- Stripe >> Ok



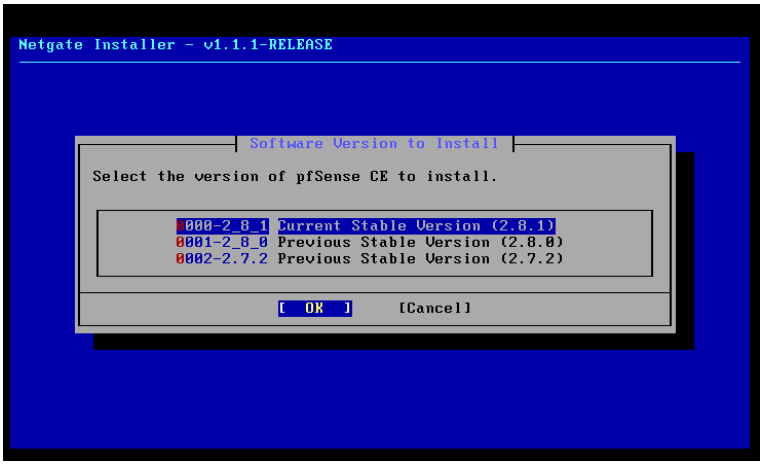
- Disk Selection >> da0 >> Ok



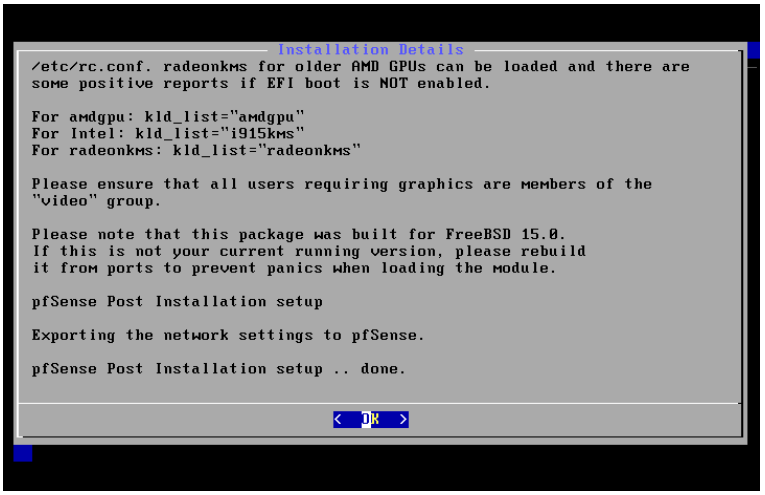
- Confirmation >> Yes if disk selection is correct.



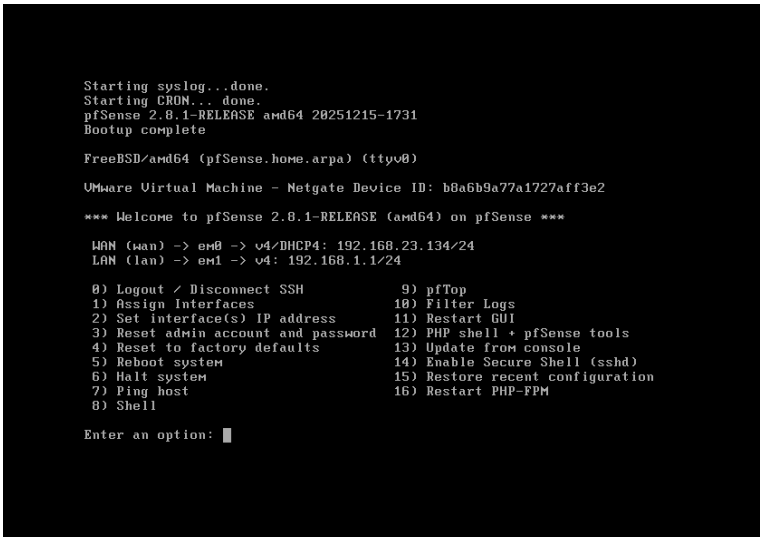
- Software Version to install
- Choose Stable and latest when in my case (2.8.1) > Ok



- Installation Details >> Ok

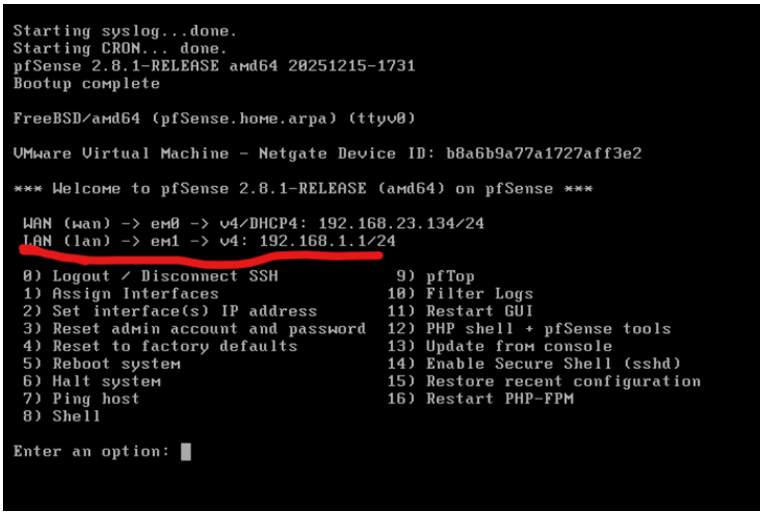


- hle

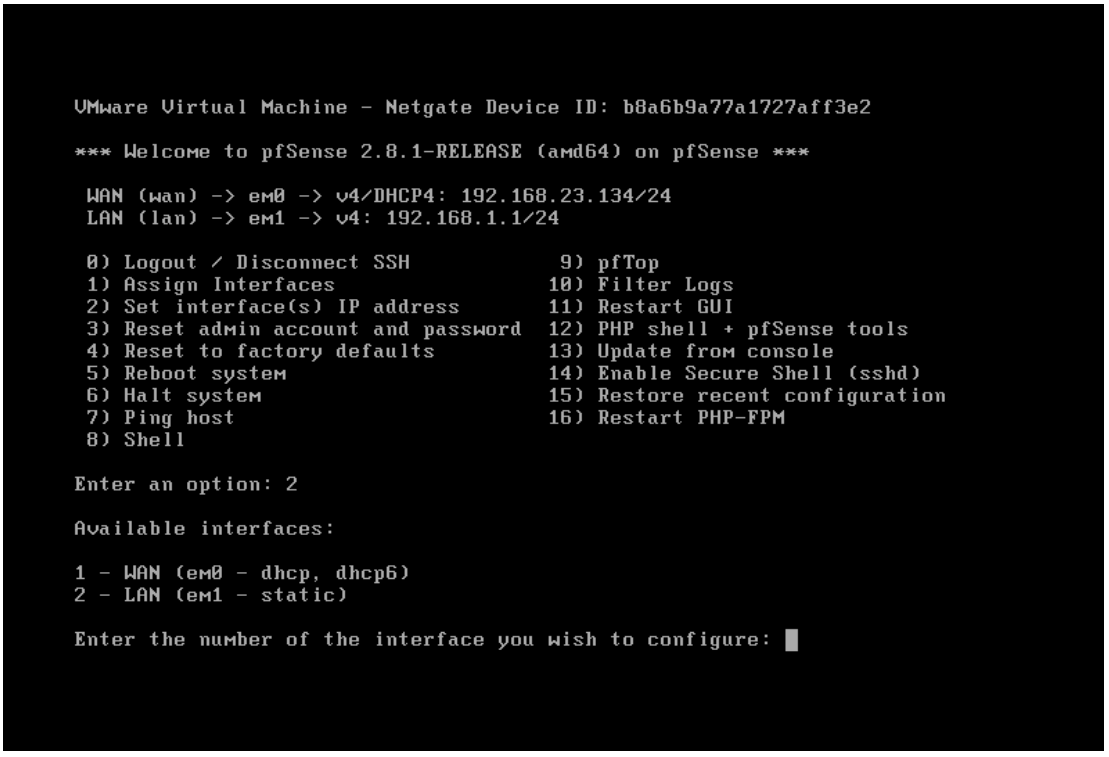


▼ Configuration

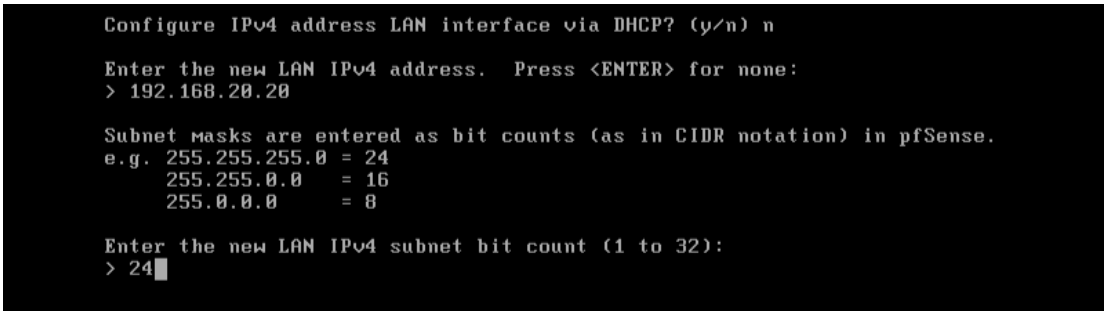
- LAN is not configured because the IP here doesn’t match with the one we assign creating Host only Adapter.
- Choose **Option 2)** Set Interface(s) IP address



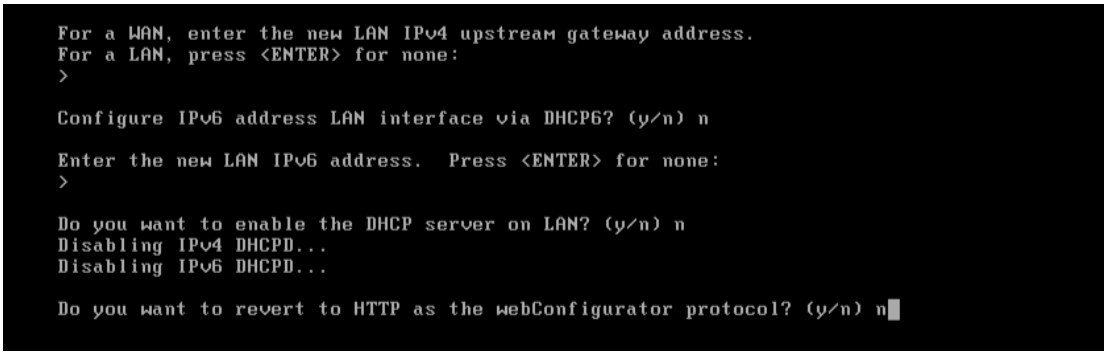
- Interface you want to configure >> 2 (LAN)



- DHCP is disable so >> Configure IPv4 address LAN interface via DHCP? >> n
- >>Enter new IPv4 address >> 192.168.20.20 >> subnet Mask >> 24
- **Note:** IPv4 address and subnet mask changes according to setting you selected during Virtual Adapter (Host Only) setting.



- Since we are configuring for LAN >> Press <Enter>
- Configure IPv6 address LAN interface via DHCP6 >> n
- Enter the new LAN IPv6 address >> n
- Do you want to revert to HTTP as the webConfigurator protocol >> n

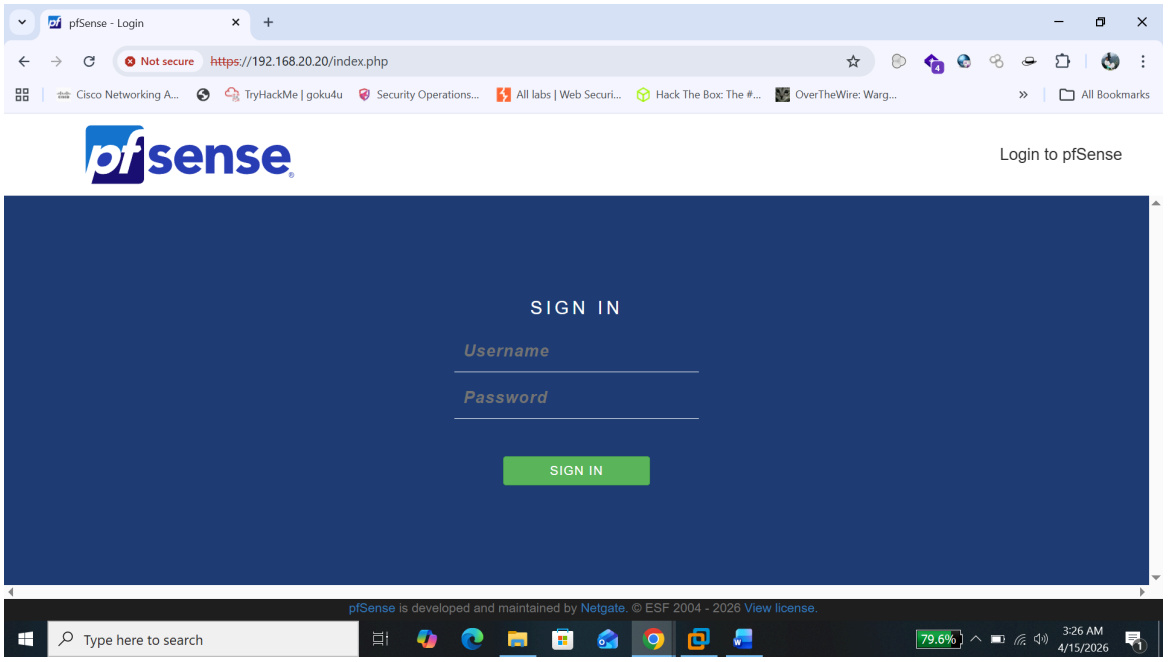


- After this installation will be done and the system will reboot.

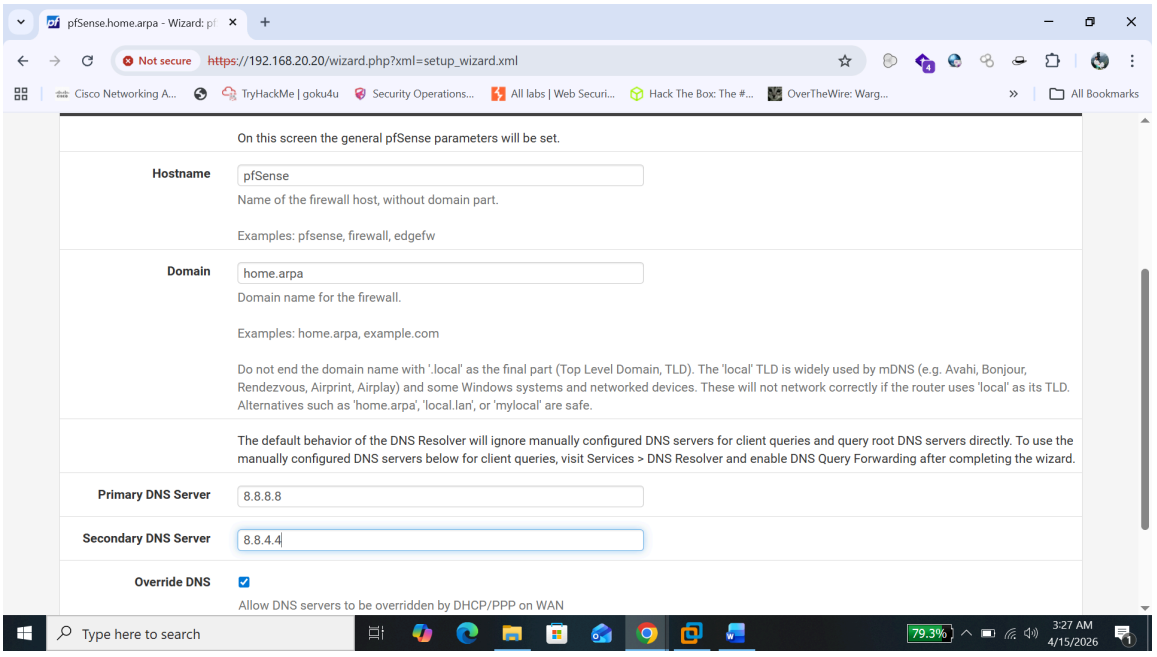
  during that phase remove the ISO file or shutdown it remove and then start.

Graphical Configuration

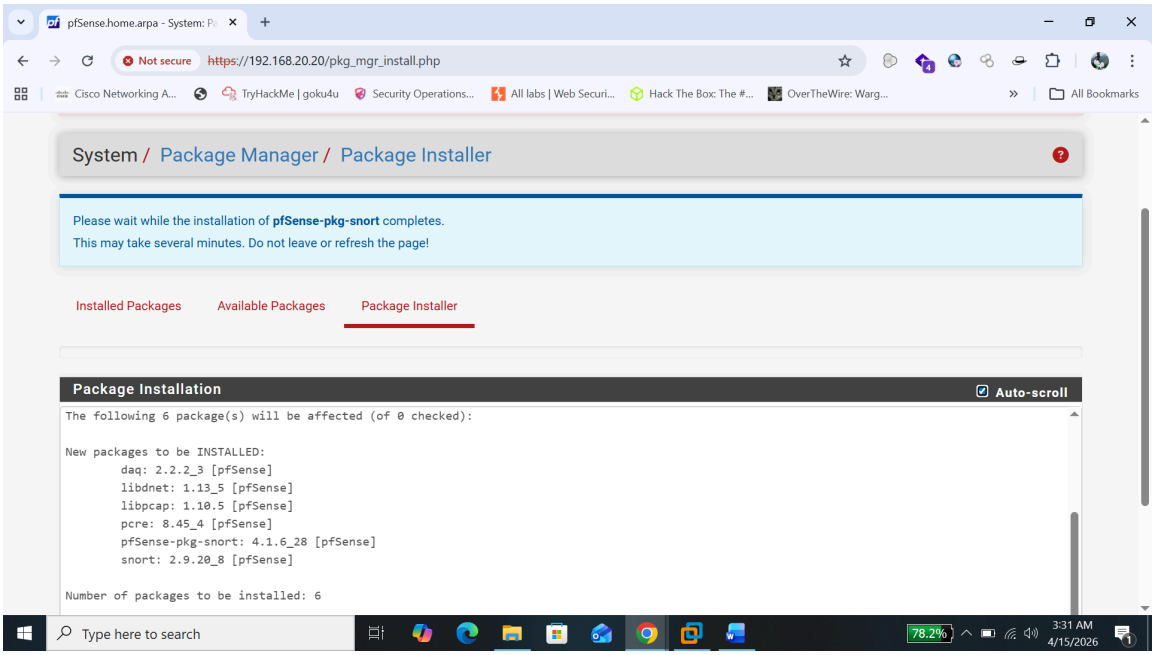
- Access the pfsense through browser using the interface ip that we assign (192.168.20.20)



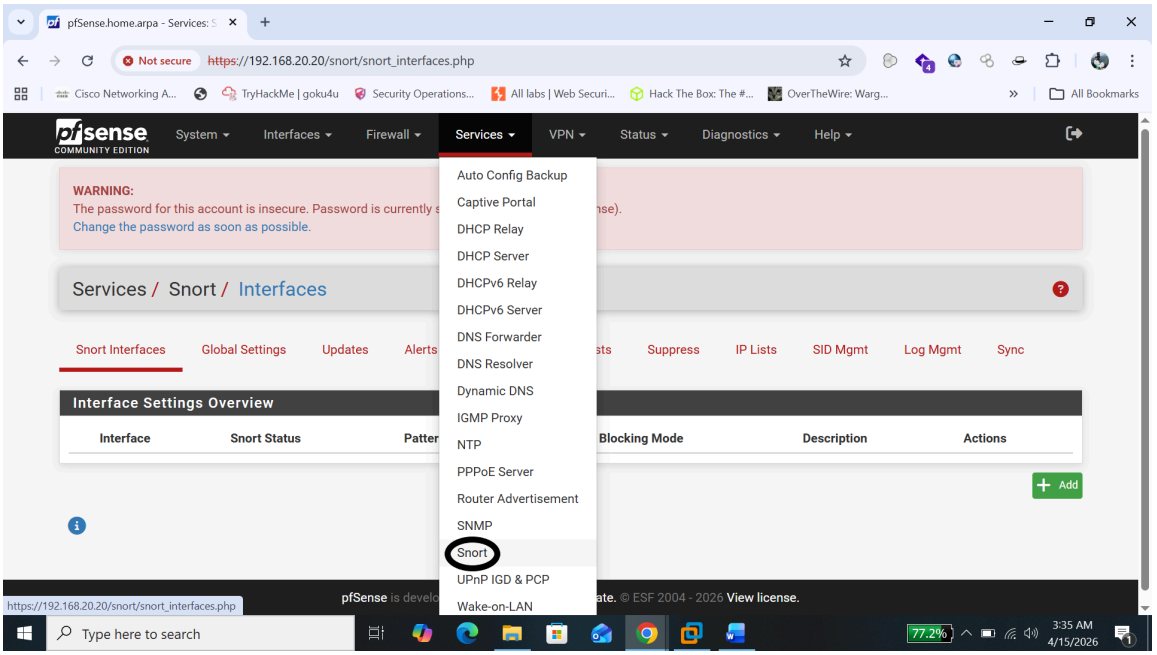
- Basic Setup
- DNS Allocation



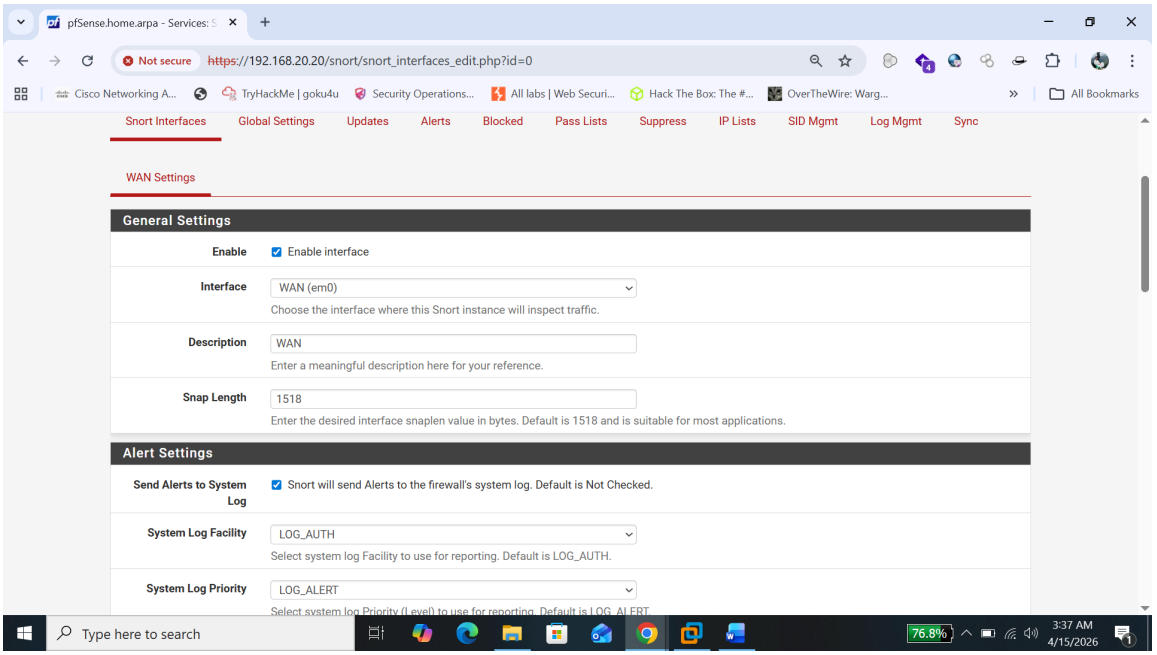
- Goto >> System >> Package Manager >> Available Packages
- Search Bar (Snort)
- Install the package checked it in Installed Packages.



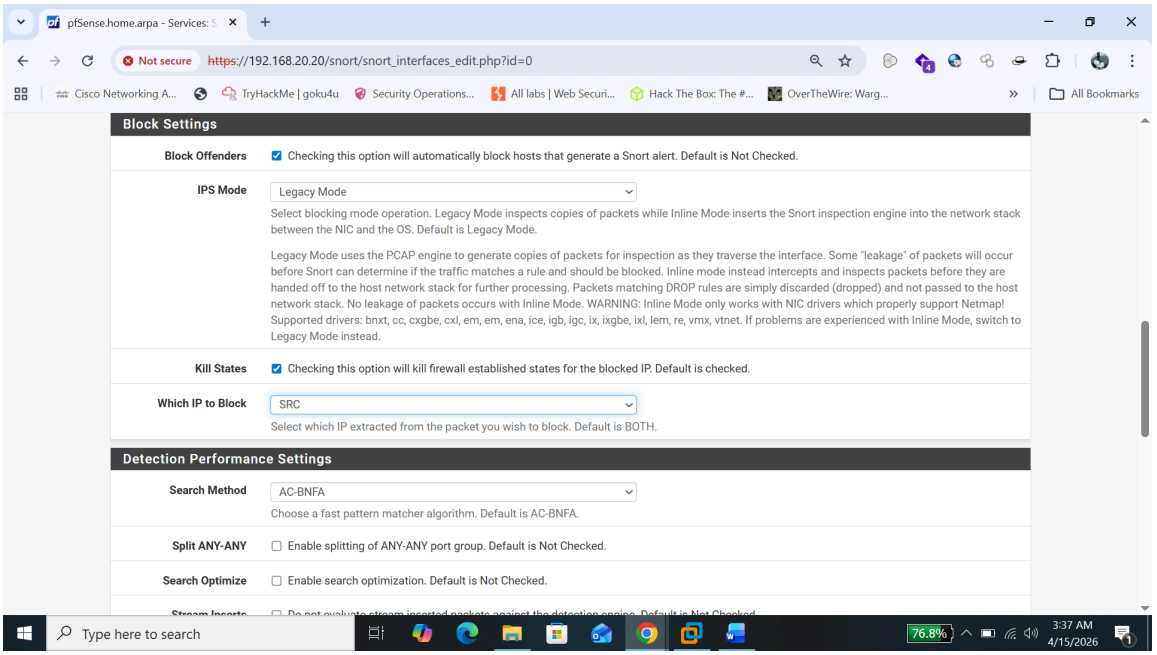
- >> Services >> Snort >> Interfaces



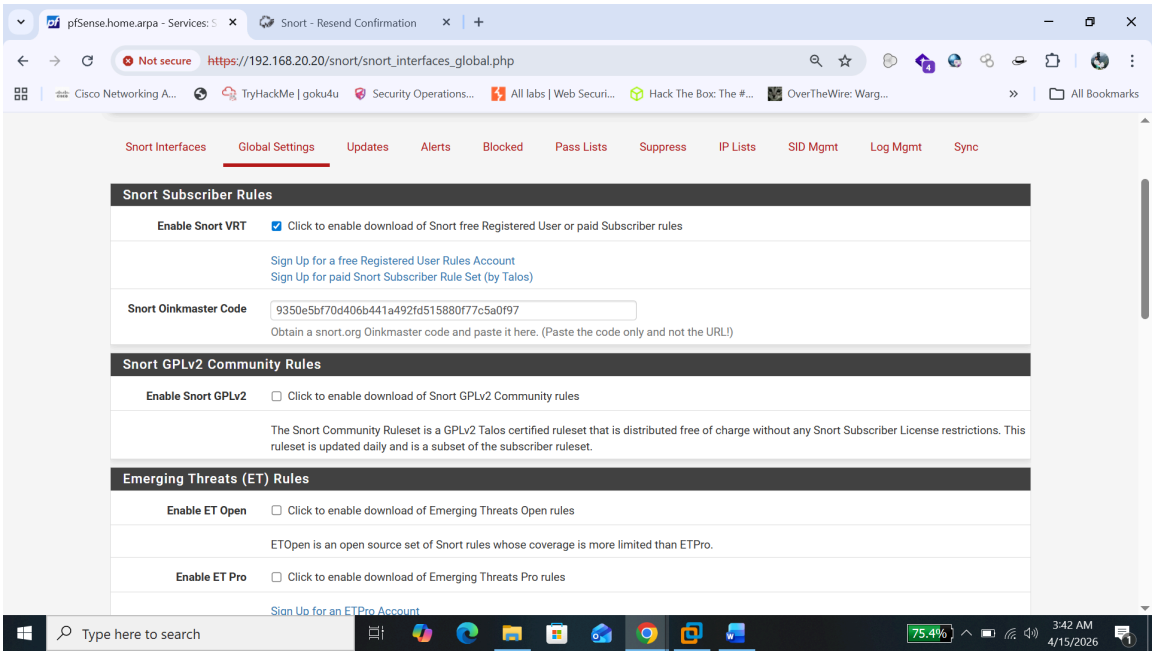
- >> WAN Setting



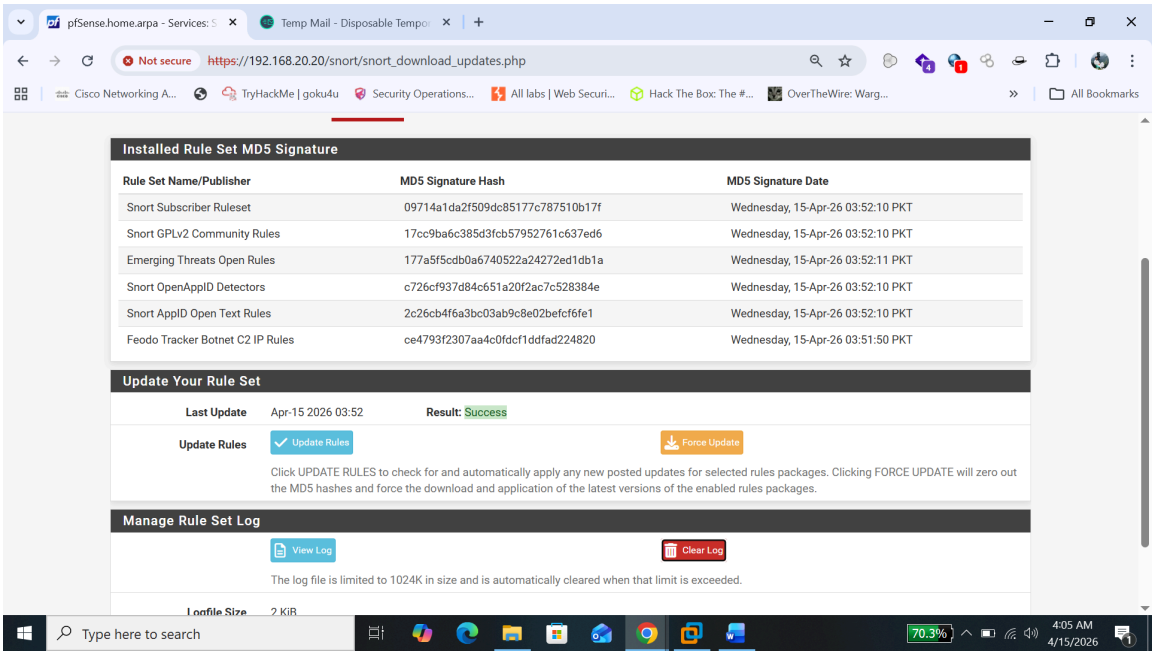
- Block settings >> Detection Performance Settings



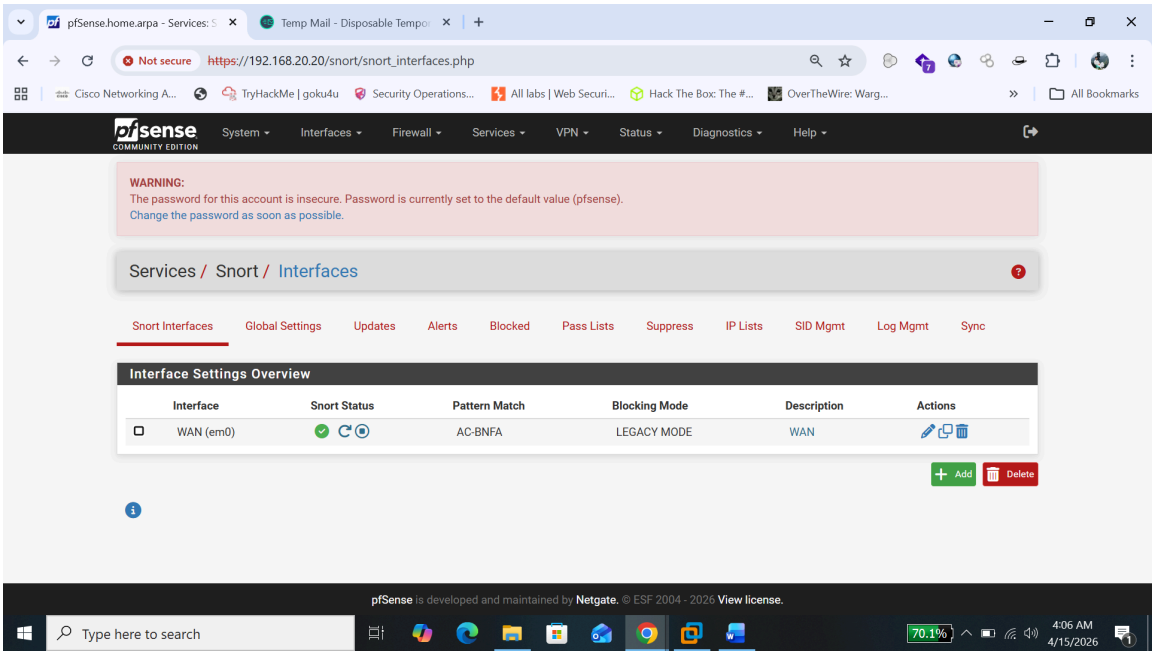
- Snort Subscriber Rules >> signup >> create account >> Copy Snort Oinkmaster Code
- Snort GPLv2 Community Rules
- Emerging Threats Rules



- Update Rules

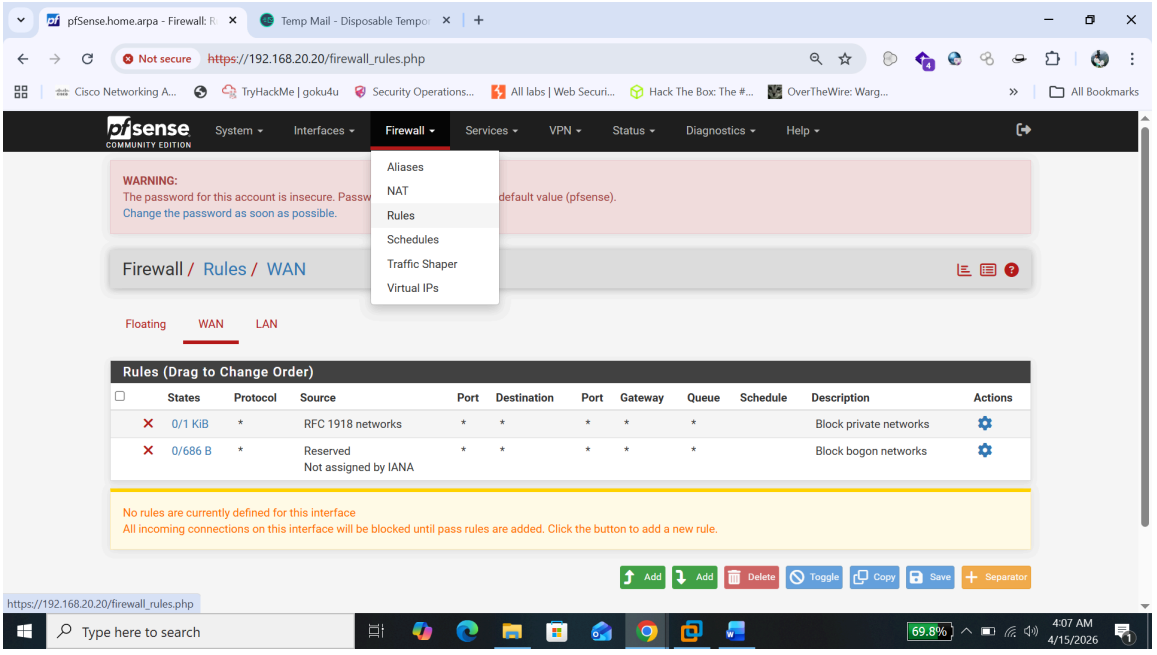


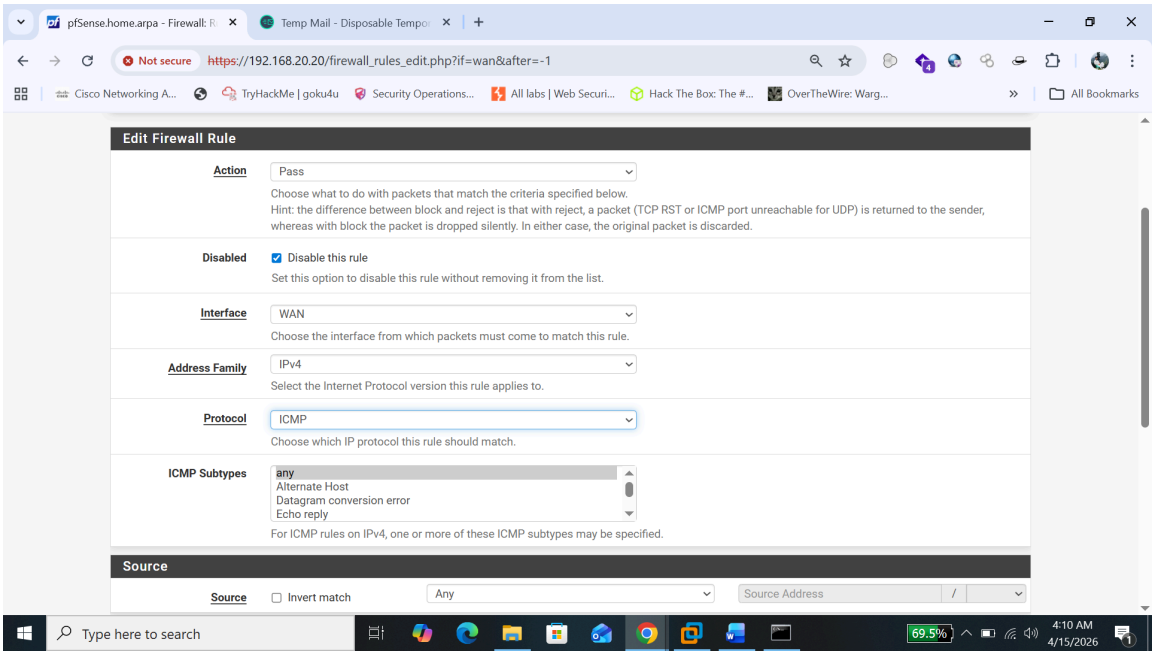
- Snort Interface >> WAN (em0) >> Start



- Remove the Default Rules (Drag to change Order)

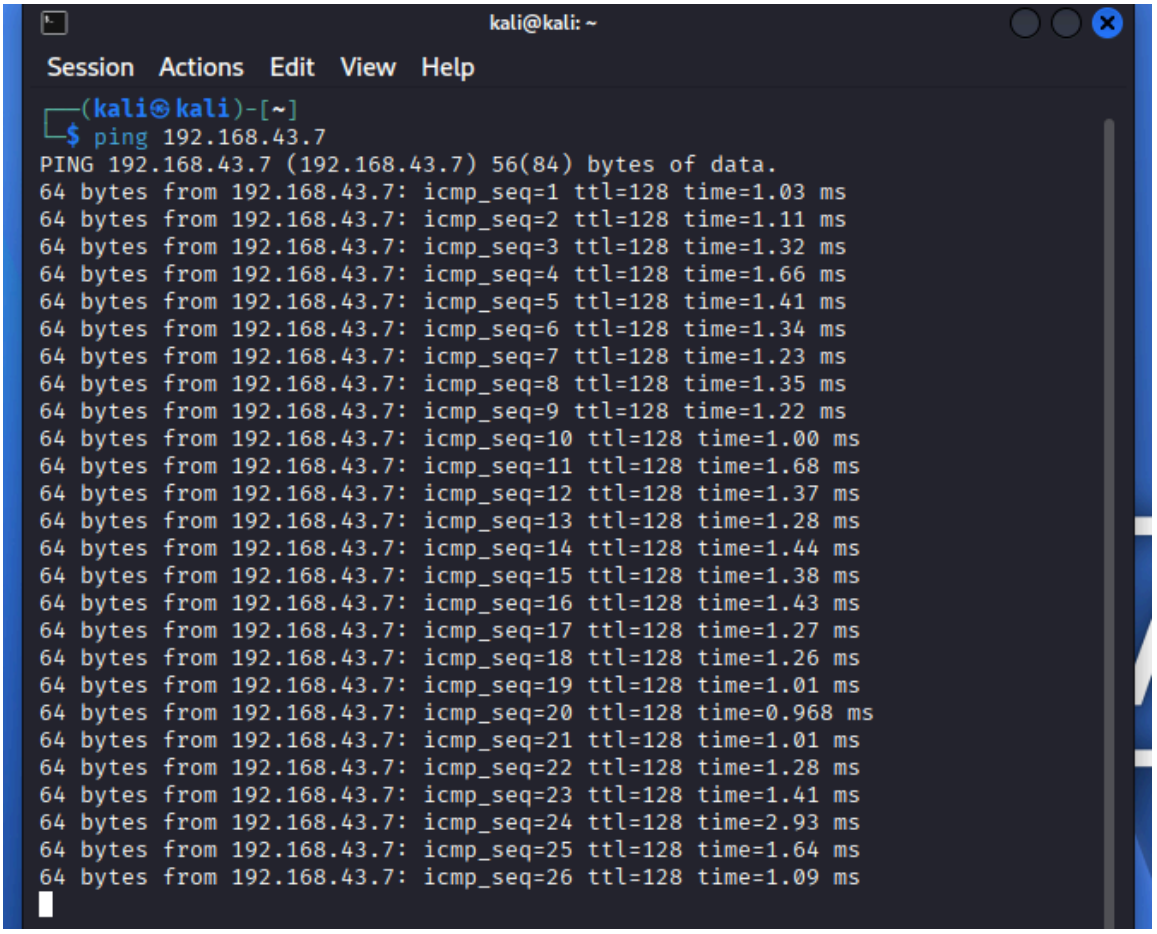
  Block private networks
- Firewall >> Rules >> WAN
- ADD >> Set for ICMP packet allow





  - Uncheck the **Disable** option (Disable this rule)

- I was facing issue with icmp packet being blocked so i have to delete the (Block private network, bogon network.
- Packet Send from machine.



- IP on Pfsense to which we send ICMP packet WAN.

